

NMAP SCAN REPORT

Network Security & Scanning — Task 2

1. Objective

The objective of this Nmap assessment was to identify active ports, network services, service versions, and the operating system of the authorized Metasploitable2 test machine.

2. Lab Environment

Attacker/Test Machine: Kali Linux

Target: Metasploitable2

Target IP: 192.168.56.101

Tool: Nmap

All scanning was performed within the authorized private virtual lab environment.

3. Host Discovery — Ping Sweep

Command

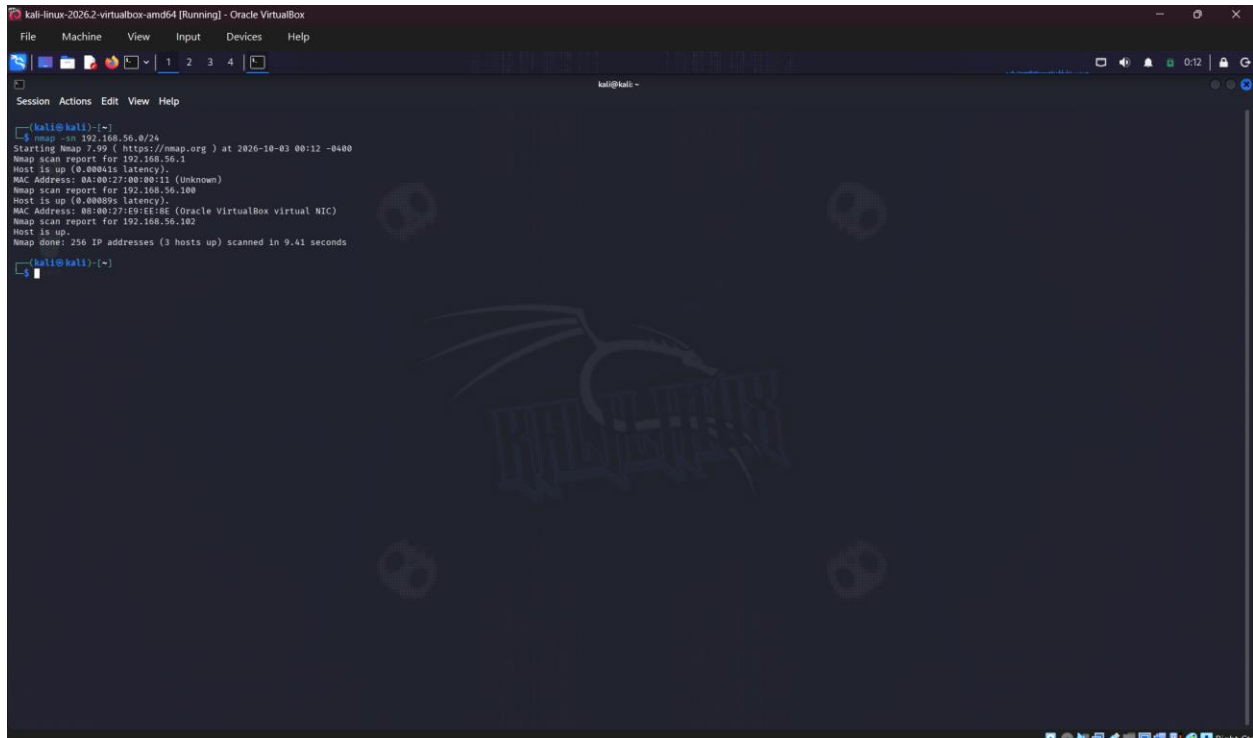
```
nmap -sn 192.168.56.0/24
```

Purpose

The `-sn` option performs host discovery without performing a port scan. It is used to identify active hosts on the local lab network.

Result

The scan identified active hosts within the 192.168.56.0/24 network, including the Metasploitable2 target.



4. TCP SYN Scan

Command

```
nmap -sS 192.168.56.101
```

Purpose

The TCP SYN scan (-sS) checks TCP ports using SYN packets and identifies accessible TCP services.

Result

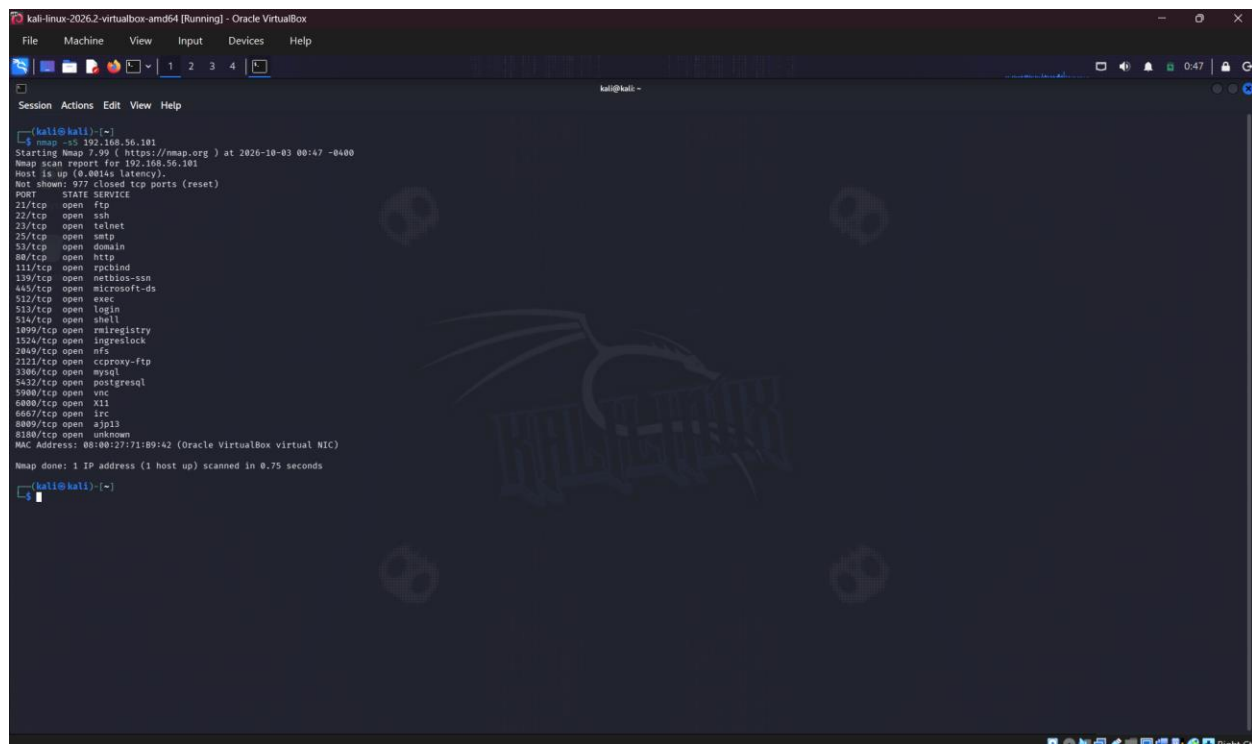
The scan identified **23 open TCP ports** on Metasploitable2.

Port	State	Service
21	Open	FTP
22	Open	SSH

23	Open	Telnet
25	Open	SMTP
53	Open	DNS
80	Open	HTTP
111	Open	RPC
139	Open	NetBIOS
445	Open	Microsoft-DS/SMB
512	Open	exec
513	Open	login
514	Open	shell
1099	Open	rmiregistry
1524	Open	ingreslock/root shell
2049	Open	NFS
2121	Open	FTP
3306	Open	MySQL
5432	Open	PostgreSQL
5900	Open	VNC
6000	Open	X11
6667	Open	IRC
8009	Open	AJP13
8180	Open	HTTP

Security Observation

The target exposes a large number of network services. Each exposed service represents part of the system's network attack surface.



```
kali@kali:~$ nmap -sS 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 00:47 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  microsoft-ds
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1899/tcp  open  maildir
1524/tcp  open  imaps
2049/tcp  open  nfs
2223/tcp  open  cifs
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:71:B9:42 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds
kali@kali:~$
```

5. UDP Scan

Command

`nmap -sU 192.168.56.101`

Purpose

The `-sU` option scans UDP ports. UDP scanning is important because services can operate over UDP even when they are not visible in a TCP scan.

Result

The scan reported:

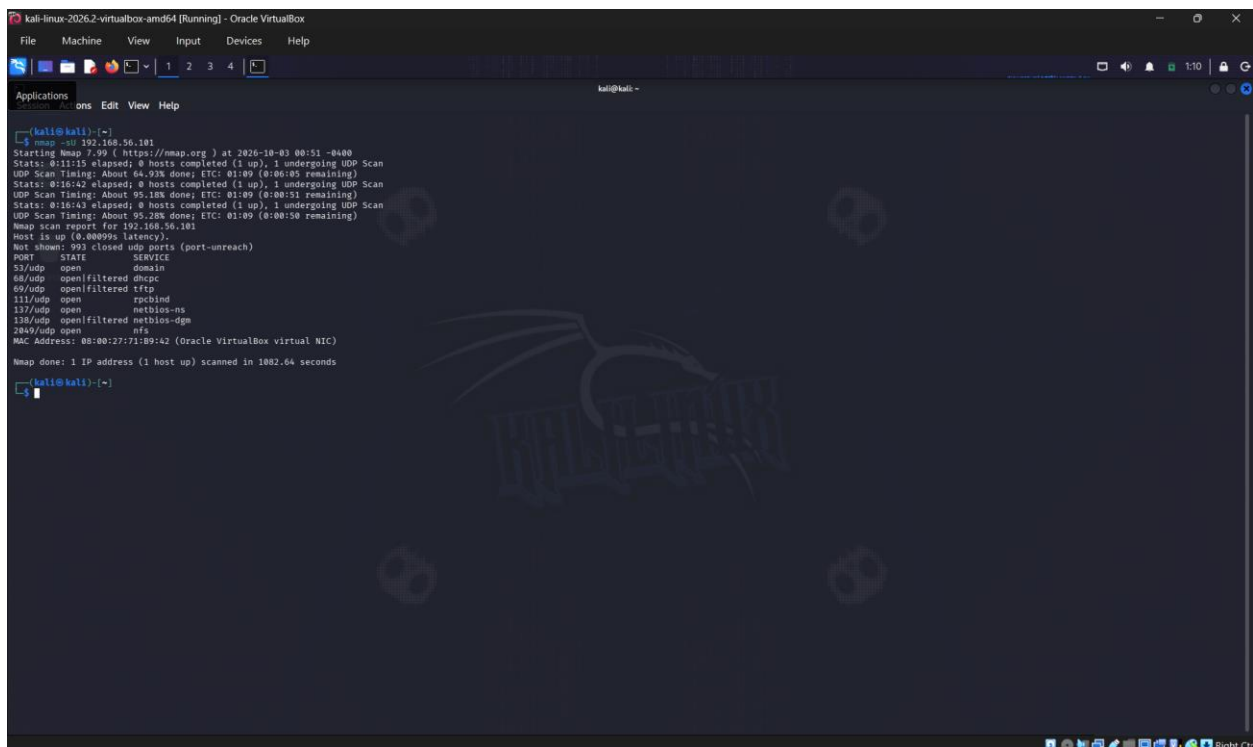
Port	State	Service
53	Open	DNS
68	Open/Filtered	DHCP
69	Open/Filtered	TFTP
111	Open/Filtered	RPC

137	Open	NetBIOS
138	Open/Filtered	NetBIOS
2049	Open/Filtered	NFS

Nmap also reported **993 closed UDP ports** with port -unreachable responses.

Security Observation

UDP services should be included in network assessments because TCP scanning alone does not provide a complete picture of network exposure.



```
kali@kali:~$ nmap -sU 192.168.56.101
Starting Nmap 7.99 (https://nmap.org) at 2026-10-03 00:51 -0400
Stats: 0:11:15 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 64.93% done; ETC: 01:09 (0:06:05 remaining)
Stats: 0:16:42 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 95.18% done; ETC: 01:09 (0:00:51 remaining)
Stats: 0:16:43 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 95.28% done; ETC: 01:09 (0:00:50 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.00099s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open/filtered dhcp
69/udp    open/filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open/filtered netbios-dgm
2049/udp  open       nfs
Nmap done: 1 IP address (1 host up) scanned in 1002.64 seconds
kali@kali:~$
```

6. Service and Version Detection

Command

```
nmap -sV 192.168.56.101
```

Purpose

The -sV option attempts to determine the service and software version running on each discovered port.

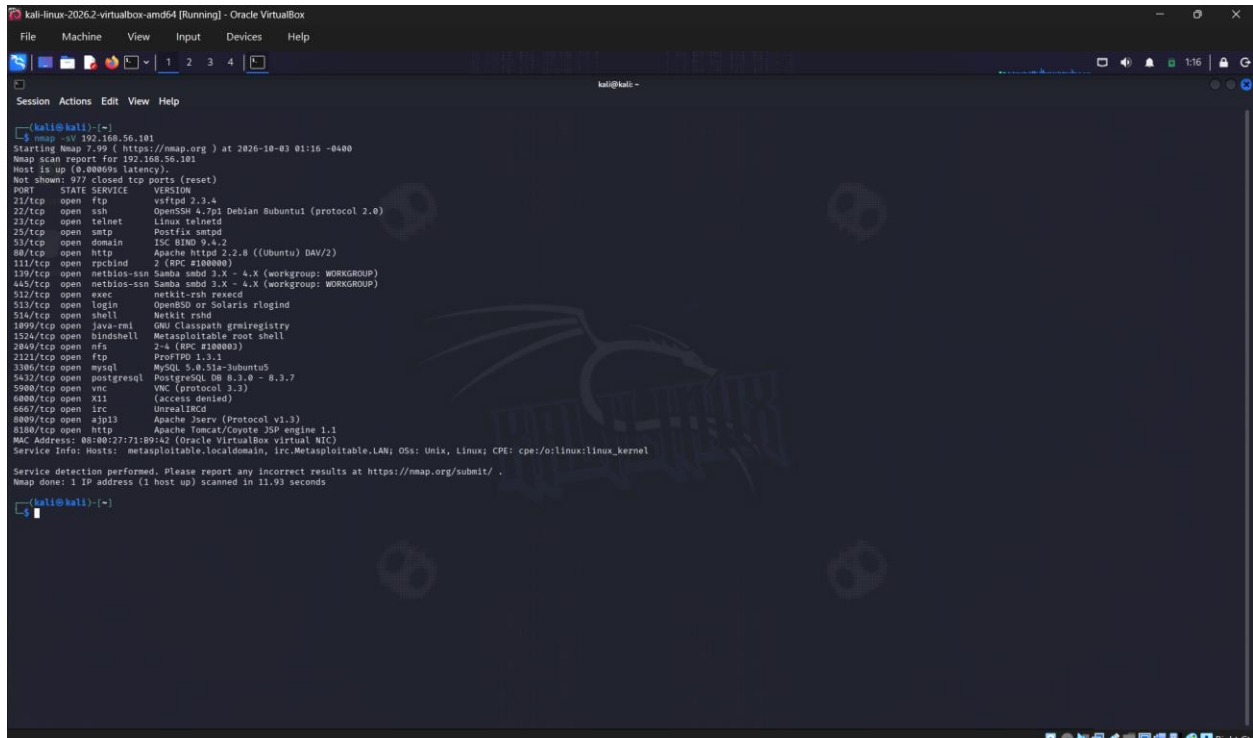
Results

Important identified services included:

Port	Service / Version
21	vsftpd 2.3.4
22	OpenSSH 4.7p1
23	Linux telnetd
25	Postfix smtpd
53	ISC BIND 9.4.2
80	Apache httpd 2.2.8
111	RPC
139	Samba smbd 3.X-4.X
445	Samba smbd 3.X-4.X
512	netkit-rsh rexec
513	OpenBSD/Solaris rlogin
514	Netkit rshd
1099	GNU Classpath grmiregistry
1524	Metasploitable root shell
2049	NFS
2121	ProFTPD 1.3.1
3306	MySQL 5.0.51a
5432	PostgreSQL 8.3.x
5900	VNC
6000	X11
6667	UnrealIRCd
8009	Apache JServ
8180	Apache Tomcat/Coyote

Security Observation

Service-version information provides useful information for vulnerability assessment and security auditing. Older software versions may require additional investigation for known vulnerabilities.



```
kali@kali:~$ nmap -v 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 01:16 -0400
Nmap scan report for 192.168.56.101
Host is up (0.000099s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian Buntul (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
135/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login          OpenSSH or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmlregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.6.51a-subuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5980/tcp  open  vnc            VNC (protocol 3.3) (access denied)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8080/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:71:B9:42 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.93 seconds
```

7. Operating System Detection

Command

```
nmap -O 192.168.56.101
```

Purpose

The -O option attempts to identify the operating system based on network characteristics.

Result

Nmap reported:

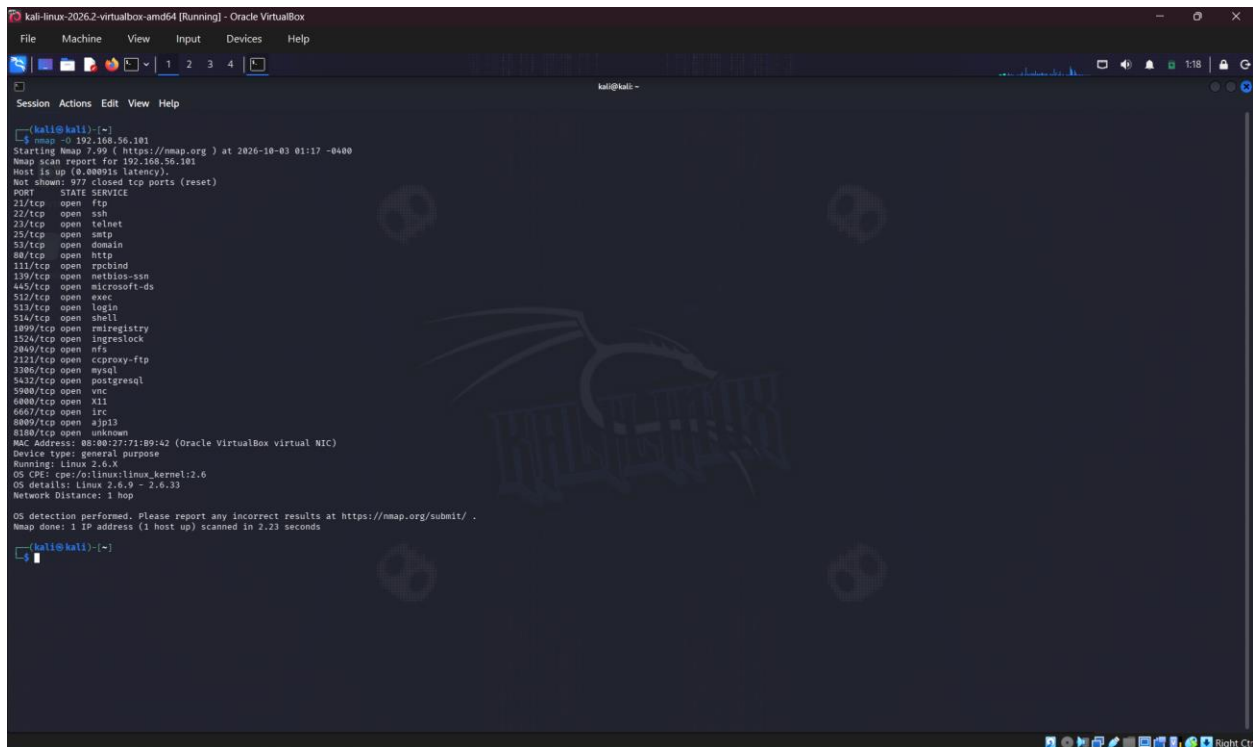
Running: Linux 2.6.X

OS details: Linux 2.6.9 - 2.6.33

Network Distance: 1 hop

Security Observation

OS identification helps security testers understand the target environment and determine which security assessments may be relevant.



```
(kali@kali):~$ nmap -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 01:17 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00001s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3206/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:71:B9:42 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.23 seconds
(kali@kali):~$
```

8. Nmap Scan Summary

Scan	Command	Purpose
Host Discovery	<code>nmap -sn</code>	Identify active hosts
TCP SYN	<code>nmap -sS</code>	Identify open TCP ports

UDP	<code>nmap -sU</code>	Identify UDP services
Service Detection	<code>nmap -sV</code>	Identify services and versions
OS Detection	<code>nmap -O</code>	Identify probable operating system

9. Key Findings

1. The Metasploitable2 machine exposed **23 TCP ports**.
2. Multiple UDP services were also detected.
3. Several services revealed specific software versions.
4. The target was identified as running an older Linux 2.6.x environment.
5. The large number of exposed services demonstrates the importance of reducing unnecessary network exposure.
6. Service-version detection provides information that can be used during subsequent vulnerability assessment.

10. Conclusion

The Nmap assessment successfully demonstrated host discovery, TCP and UDP port scanning, service/version detection, and operating-system detection against the authorized Metasploitable2 laboratory system.

The results provide a baseline view of the target's network exposure and demonstrate how Nmap can be used as an important reconnaissance and network-security assessment tool.